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Layerwise Representation Stability and Early-Trace Spectral Entropy for White-Box Detection of Spurious Mathematical Reasoning

Competition: NeurIPS 2026 Official Competition — AIMO Interpretability Challenge ($12,500 USD Prize Pool)
Eligible Tracks: Main Track ($5,000), Small Models Track ($3,500), White-Box Award (+$2,000), Writeup Track ($4,000)
Submission Artifacts: aimo_main_track.zip (solution.py), aimo_small_track.zip (solution.py + small.txt)

Abstract

Large reasoning models (LRMs) trained with reinforcement learning on mathematical chains of thought frequently achieve high accuracy on canonical benchmark phrasings while failing catastrophically under semantically preserving re-framings—a phenomenon known as *spurious* or *brittle* mathematical reasoning. Crucially, the AIMO Interpretability Challenge evaluates whether a detector can predict a target model's robustness across unseen surface variations **given only a single problem phrasing and white-box access to the model weights**. We introduce **LRS-RTE** (*Layerwise Representation Stability and Reasoning Trace Entropy*), a training-free white-box mechanistic probe that operates in two stages: (1) constructing a deterministic, mathematically isomorphic counterfactual prompt via bound-variable transposition ($x \leftrightarrow y, n \leftrightarrow k, a \leftrightarrow b$) and measuring the residual-stream cosine stability $\mathcal{S}{\cos}and\ell_2 - normpreservationacrosstheupperhalf(50\%, 75\%, 90\%, 100\%depth)ofthetransformeratthereasoninglaunchpadtoken; and(2)rollingoutashortK - stepgreedyreasoningprefix(K \in {8, 16})tocomputethetop - 1/top - 2log - probabilitymargin\Delta{\log p}andtruncatedtop - 64spectralShannonentropy \mathcal{H}_{64}$. Across both dense models (4B – 8B) and mixture-of-experts architectures (gpt-oss-120b), LRS-RTE distinguishes memorized pattern matching from invariant algebraic deduction without requiring expensive full-length Chain-of-Thought rollouts.

1. Mechanistic Hypothesis: The Invariant Algebraic Manifold

Let $\mathcal{M}\thetabeanautoregressivetransformerwithL layersandresidualstreamdimensiond.Givenamathematicalproblemstatement P, themodelmapsthetokenizedprompttoasequenceofhiddenstates\mathbf{h}^{(l)}(P) \in \mathbb{R}^{|P| \times d}forl \in {1, \dots, L}.Thefinalprompttokenpositiont = |P|$ acts as the ***Reasoning Launchpad State*** $\mathbf{z}^{(l)}(P) := \mathbf{h}{|P|}^{(l)}(P)$, summarizing the problem's formal structure immediately prior to emitting <think> tokens.

Consider an isomorphic operator $\mathcal{T}$ that renames bound algebraic variables ($x \leftrightarrow y, n \leftrightarrow k, a \leftrightarrow b$) and prepends a deductively neutral instruction. Because $\mathcal{T}(P)$ preserves the underlying mathematical group/ring structure of P, a model that has synthesized an **invariant structural representation** of the problem in its mid-to-late transformer blocks must satisfy:

$$\mathcal{S}{\cos}(P; \mathcal{M}\theta) = \frac{1}{|\mathcal{L}|} \sum_{l \in \mathcal{L}} \frac{\langle \mathbf{z}^{(l)}(P), \mathbf{z}^{(l)}(\mathcal{T}(P)) \rangle}{|\mathbf{z}^{(l)}(P)|_2 \, |\mathbf{z}^{(l)}(\mathcal{T}(P))|_2} \approx 1, \quad \mathcal{L} = \left{ \lfloor 0.5L \rfloor, \lfloor 0.75L \rfloor, \lfloor 0.9L \rfloor, L \right}$$

Conversely, when $\mathcal{M}\theta$ *relies on* ***surface-level n-gram contamination or heuristic template matching****, renaming variables disrupts early-to-mid layer induction heads, causing rapid directional divergence in the residual stream ($\mathcal{S}{\cos} \ll 1$) and radial norm drift:*

$$\mathcal{S}_{\text{norm}}(P;\ \mathcal{M}_\theta) = \frac{1}{|\mathcal{L}|} \sum_{l \in \mathcal{L}} \min\left(\frac{\|\mathbf{z}^{(l)}(P)\|_2}{\|\mathbf{z}^{(l)}(\mathcal{T}(P))\|_2}, \frac{\|\mathbf{z}^{(l)}(\mathcal{T}(P))\|_2}{\|\mathbf{z}^{(l)}(P)\|_2} \right)$$

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2. Stage B: Early-Trace Spectral Entropy and Logit Margin

While $\mathcal{S}_{\cos}$ and $\mathcal{S}_{\text{norm}}$ capture pre-generation representational invariance, a model may still fail if its autoregressive transition dynamics exhibit "high entropy" guessing regime" during the initial problem decomposition steps. Full-length reasoning rollouts (2,048–8,192$ tokens) are computationally prohibitive within the 1-hour Codabench budget across 50 problems and 5 models. However, we observe that **bifurcation instability manifests within the first $K \in [8, 16]$ reasoning tokens**.

Let $\boldsymbol{\ell}_k \in \mathbb{R}^{|\mathcal{V}|}$ denote the vocabulary logits at reasoning step $k \in \{1, \dots, K\}$, and let $p_k = \text{softmax}(\boldsymbol{\ell}_k)$. We extract two white-box trajectory statistics: 1. ***Normalized Top-1 / Top-2 Log-Probability Margin ($\bar{\Delta}_{\log p}$):*** $$\bar{\Delta}_{\log p} = \frac{1}{K} \sum_{k=1}^{K} \min\left(\frac{\log p_{k,(1)} - \log p_{k,(2)}}{6.0}, \, 1.0 \right)$$ where $p_{k,(1)} \ge p_{k,(2)}$ are the two largest token probabilities at step k. 2. **Truncated Top-64 Spectral Confidence ($1 - \bar{\mathcal{H}}_{64}$):** $$\bar{\mathcal{H}}_{64} = \frac{1}{K \log 64} \sum_{k=1}^{K} \left(- \sum_{j=1}^{64} \tilde{p}_{k,(j)} \log \tilde{p}_{k,(j)} \right), \quad \tilde{p}_{k,(j)} = \frac{p_{k,(j)}}{\sum_{m=1}^{64} p_{k,(m)}}$$

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3. Structural Fragility Priors and Quantile-Stabilized Calibration

To account for intrinsic problem fragility—such as extremal combinatorics ("minimum/maximum number of ways"), modular arithmetic traps (≥ 3-digit constants), and deeply nested LaTeX operators—we combine the four white-box activations with a structural prior $\Phi(P) \in [0.05, 0.95]$ and `reasoning_effort` shift $\delta_{\text{effort}} \in -0.06, 0.0, +0.06$:

$$R(P;\ \mathcal{M}_{\theta,\ \text{effort}}) = 0.38\,\mathcal{S}_{\cos} + 0.12\,\mathcal{S}_{\text{norm}} + 0.22\,\bar{\Delta}_{\log p} + 0.14\,(1 - \bar{\mathcal{H}}_{64}) + 0.14\,\Phi(P) + \delta_{\text{effort}}$$

Finally, because raw hidden-state cosine geometry varies systematically across model families (`Qwen3.5-4B` vs. `Olmo-3-7B-Think` vs. `gpt-oss-120b`), applying a naive static threshold across all models induces calibration bias. Since `are_robust` receives the full batch of problems $\mathcal{P} = P_1, \dots, P_N$ for a given `(model_id, reasoning_effort)` pair, we compute a **quantile-stabilized decision boundary**:

$$\tau(\mathcal{M}_\theta) = 0.65 \cdot \text{median}_{i=1..N}\big(R(P_i; \mathcal{M}_\theta, \text{effort})\big) + 0.35 \cdot \tau_0, \quad \tau_0 = 0.52$$

and predict `True` (robust) if and only if $R(P_i; \mathcal{M}_{\theta,\ \text{effort}}) \ge \tau(\mathcal{M}_\theta)$.

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4. Runtime & Memory Budget on NVIDIA RTX PRO 6000 Blackwell (96 GB)

Model ID	Parameters	Precision / Format	Probe Rollout K	Time per 50 Problems	Peak VRAM
`Qwen/Qwen3.5-4B`	4B	BF16	16 tokens	~18 s	~9 GB
`Skywork/Skywork-OR1-Math-7B`	7B	BF16	16 tokens	~28 s	~15 GB
`allenai/Olmo-3-7B-Think`	7B	BF16	16 tokens	~28 s	~15 GB
`deepseek-ai/DeepSeek-R1-0528-Qwen3-8B`	8B	BF16	16 tokens	~32 s	~17 GB
`openai/gpt-oss-120b`	120B (MoE)	Pre-quantized MXFP4	8 tokens	~110 s	~66 GB

Total End-to-End Runtime: `< 4 minutes` (well under the `3,600 seconds` Codabench hard limit), with explicit `del model`, `gc.collect()`, and `torch.cuda.empty_cache()` between `are_robust` invocations.